GreenNirman

India's Green Building Assessment Platform

Empowering Sustainable Construction Decisions Across All 758 Indian Districts

Complete Technical & Data Summary | March 2026

This document provides a complete technical summary of GreenNirman — India's district-level green building assessment platform. It covers data sources, geographic coverage, climate methodology, materials database, calculation standards, and the platform's role in advancing sustainable construction across India.

1. Vision & Purpose

India's building sector consumes approximately 40% of national energy and contributes ~22% of total CO2 emissions. With over 9.2 million new buildings constructed annually — projected to reach 11.3 million by 2028 — and rapid urbanisation continuing, the decisions made at the design stage have a profound and lasting impact on India's carbon footprint, water resources, and energy demand.

GreenNirman was built to address a critical gap: making evidence-based, India-specific green building assessment accessible at the district level — not just in major metros. Every district in India has a different climate, different soil conditions, different seismic risk, different locally available materials, and different water stress. Generic national-level tools cannot capture this complexity.

What GreenNirman Does

- Assesses the sustainability of a building design before construction begins
- Provides district-specific climate data, material recommendations, and compliance checks
- Calculates energy savings, carbon reduction, and lifecycle cost for any building in India
- Checks compliance with ECBC 2017, Eco Niwas Samhita 2021, and GRIHA rating criteria
- Generates a professional PDF assessment report for authorities, lenders, and planners

Who Benefits

Stakeholder	How GreenNirman Helps	Impact
Architects & Civil Engineers	Instant ECBC/GRIHA compliance check during design phase	Reduces rework, ensures code compliance from day one
Urban Local Bodies / Municipalities	Standardised green building assessment for approvals	Faster, data-driven green building permissions
Housing Boards & Developers	Material comparison with cost and carbon data	Informed decisions that reduce lifecycle cost 15-25%
Individual Homeowners	Understand energy savings and water risk before building	Make climate-smart choices specific to their district
Policy Makers & Planners	District-level sustainability baseline data	Evidence for local green building bylaws and schemes
Research & Academia	Open climate + material dataset for 758 districts	India-specific embodied carbon and energy research

2. India's Sustainability Challenge

India has committed to reducing emissions intensity of GDP by 45% by 2030 (NDC update 2022) and achieving net zero by 2070. The built environment is central to meeting these targets.

Key Facts — India's Built Environment

Indicator	Value	Source
Share of national energy consumption	~40%	BEE, MoEFCC
Share of CO2 emissions from buildings	~22%	GRIHA Council
New buildings constructed annually	~9.2 million (2023)	NBO Housing Statistics 2023
Districts covered by ECBC compliance tools	< 50 (metros only)	BEE survey 2022
Estimated cooling energy demand growth by 2050	~9x by 2050	IEA / CLASP 2024
Urban population by 2050 (projected)	~951 million	UN World Urbanization Prospects
PMAY housing units targeted	29.5 million	MoHUA 2023

The Gap GreenNirman Fills

- Existing tools (GRIHA, LEED India, IGBC) require expensive consultants and are limited to large projects
- No accessible tool covers all 758 Indian districts with real climate data
- Most builders and homeowners in tier-2/tier-3 cities have no access to green building guidance
- ECBC 2017 mandates compliance for buildings >500m² but enforcement is limited by lack of accessible tools
- India's diverse climate zones (5 zones, from Leh to Lakshadweep) require district-level precision, not national averages

3. Geographic Data Coverage

GreenNirman contains verified data for all 758 official Indian districts across 36 states and union territories — making it the most geographically comprehensive India-specific green building tool available.

Three District Data Tables

Data Table	Districts	Coverage	What It Contains	Source
DISTRICT_ZONE	758	100%	ECBC Climate Zone 1-5 for every district	BEE / ECBC 2017
DISTRICT_CLIMATE_DB	670	88%	Monthly temp, humidity, solar — 12 months, 21-year average	NASA POWER API (2000-2020)
DISTRICT_INSIGHTS	758	100%	Seismic zone, flood risk, soil type, water scarcity, construction cost, recommended materials	BIS IS1893:2016, NDMA, NBSS&LUP, CGWB 2022, CPWD DSR 2024

88 districts without individual NASA data (mostly very remote or small districts in Northeast India and island territories) use the long-term average for their ECBC climate zone. All calculations remain valid.

ECBC Climate Zones — All 5 Covered

Zone	Name	Climate Character	Example Districts	~Districts
Zone 1	Hot & Dry	High heat, low humidity, desert/semi-arid	Jaipur, Jodhpur, Jaisalmer, Hyderabad, Nagpur	~180
Zone 2	Warm & Humid	High humidity, coastal, moderate-high temp	Mumbai, Chennai, Kochi, Visakhapatnam, Mangaluru	~140
Zone 3	Composite	Dual season — hot summer + mild winter	Delhi, Lucknow, Pune, Bengaluru, Bhopal	~250
Zone 4	Temperate	Moderate climate, long natural ventilation season	Mysuru, Coimbatore, Shillong, Darjeeling	~80
Zone 5	Cold	Cold winters, snow, high altitude	Leh, Srinagar, Shimla, Dehradun, Mussoorie	~108

State-wise District Coverage

State / UT	Districts	State / UT	Districts	State / UT	Districts
Uttar Pradesh	75	Madhya Pradesh	52	Bihar	38
Tamil Nadu	37	Maharashtra	36	Assam	35
Chhattisgarh	33	Gujarat	33	Rajasthan	33
Telangana	33	Karnataka	31	Odisha	30
Arunachal Pradesh	25	West Bengal	23	Punjab	23
Jharkhand	24	Andhra Pradesh	26	Haryana	22
Jammu & Kashmir	20	Manipur	16	Kerala	14
Uttarakhand	13	Himachal Pradesh	12	Meghalaya	12
Nagaland	15	Delhi	11	Mizoram	11
Sikkim	6	Tripura	8	Puducherry	4
Andaman & Nicobar	3	Dadra & NH, D&D	3	Ladakh	2
Goa	2	Chandigarh	1	Lakshadweep	1

4. Climate Data — NASA POWER

For 670 districts, GreenNirman stores 21-year averaged climate data (2000–2020) fetched from the NASA POWER API. This real satellite-derived data directly powers all energy calculations — making results specific to each district's actual climate, not a generalised zone assumption.

Three Climate Parameters — Monthly for 12 Months

Parameter	Unit	What It Represents	Used For in Calculations
Temperature (T2M)	°C	Monthly mean air temperature at 2m height	Cooling/heating load, HDD/CDD, thermal comfort
Relative Humidity (RH2M)	%	Monthly mean relative humidity at 2m	Moisture risk, mould potential, ventilation need
Solar Radiation (ALLSKY_SFC_SW_DWN)	kWh/m²/day	All-sky surface shortwave downward irradiance	Solar heat gain (OTTV/RTTV), PV yield, passive design

How NASA POWER Data Was Collected

- NASA POWER (Prediction Of Worldwide Energy Resources) API called for each district's latitude and longitude
- 21-year monthly averages computed: 2000–2020 (252 monthly data points per district)
- Three parameters fetched: T2M (temperature), RH2M (humidity), ALLSKY_SFC_SW_DWN (solar)
- Invalid readings (value = -999) automatically excluded from averaging
- Annual HDD (Heating Degree Days) and CDD (Cooling Degree Days) calculated from monthly means
- Data stored as embedded constants — app works fully offline, no API dependency at runtime

Sample Climate Profile — Chennai (Zone 2: Warm & Humid)

Month	Temp (°C)	RH (%)	Solar (kWh/m²/day)	Design Implication
January	24.8	76	5.21	Comfortable — natural ventilation viable
February	26.3	72	6.02	Increasing heat — shading begins to matter
March	28.9	71	6.45	Solar gain significant — chajja overhang needed
April	31.2	73	6.61	High cooling load — SHGC critical

Month	Temp (°C)	RH (%)	Solar (kWh/m²/day)	Design Implication
May	33.1	70	6.38	Peak heat — SRI>100 roof strongly recommended
June	31.8	76	5.02	Monsoon onset — cross-ventilation priority
July	30.2	81	4.21	High humidity — moisture-resistant materials
August	29.8	83	4.18	Peak humidity — ventilation essential
September	29.5	82	4.67	Sustained humidity — wall breathability matters
October	28.1	81	5.11	NE monsoon — flood-resistant plinth needed
November	26.4	78	5.34	Cyclone season — wind-resistant design
December	24.9	76	5.08	Comfortable winter — minimal heating need

5. District Insights — Local Risk & Context Data

Beyond climate, every district in India has unique physical and regulatory characteristics that profoundly affect building design decisions. GreenNirman embeds 8 fields of district-level context data for all 758 districts.

Field	What It Shows	Why It Matters for Green Buildings	Source
Seismic Zone	BIS IS 1893:2016 zone (I–V)	Determines structural system requirements — affects material choice and embodied carbon	BIS / NBC 2016
Flood Risk	Low / Moderate / High / Very High	Affects foundation design, material durability selection, and plinth level requirements	NDMA Flood Atlas 2010 + NRSC 2020
Soil Type	Sandy / Black cotton / Laterite / Alluvial / Rocky	Expansive soils (black cotton) demand different foundations — impacts material and cost choices	NBSS&LUP district soil surveys
Water Scarcity	Low / Moderate / High / Critical	Determines RWH urgency, water-efficient fixture requirements, and groundwater risk	CGWB Groundwater Year Book 2022-23
Avg Construction Cost	Rs./sqft range	Contextualises lifecycle cost analysis to local market reality	CPWD DSR 2023-24 + state indices
Best Wall Materials	Top 3 climate-optimised walls	Recommends locally appropriate, climate-matched wall options from the materials database	ECBC 2017 zone guidance + local availability
Best Roof Materials	Top 3 climate-optimised roofs	Directs toward high-SRI roofs in hot zones, sloped roofs in high-rainfall zones	ECBC 2017 zone guidance
Climate Advice	District-specific design guidance	Provides actionable passive design recommendations specific to that district's conditions	NBC 2016 Part 8 + zone knowledge

District Risk Profiles — Diverse India

District	State	Seismic	Flood	Soil	Water	Cost (Rs./sqft)
Leh	Ladakh	IV	Low	Rocky / cold desert	Very Low	2,800–4,000

District	State	Seismic	Flood	Soil	Water	Cost (Rs./sqft)
Kutch	Gujarat	V	Low	Saline / sandy	Critical	1,400–1,800
Jaisalmer	Rajasthan	IV	Low	Desert sandy	Critical	1,100–1,500
Alappuzha	Kerala	III	Very High	Alluvial / backwater	Low	1,700–2,300
Chamoli	Uttarakhand	V	High	Mountain rocky / glacial	Low	1,900–2,600
Chennai	Tamil Nadu	II	High	Alluvial coastal / clay	Critical	1,800–2,500
Kamrup Metropolitan	Assam	V	High	Alluvial	Low-Moderate	1,500–2,000
South 24 Parganas	West Bengal	III	Very High	Deltaic alluvial	Moderate	1,300–1,800
Bengaluru Urban	Karnataka	II	Low	Red laterite / rocky	High	1,800–2,800
Muzaffarpur	Bihar	III	Very High	Alluvial	High	1,000–1,300

This data illustrates the enormous diversity of building conditions across India — from the permafrost-adjacent cold desert of Leh to the cyclone-prone deltaic wetlands of South 24 Parganas. A single national standard cannot address this diversity. District-level data is essential.

6. Materials Database — 72 Materials

GreenNirman's materials database contains 72 verified building materials across 5 categories, with real embodied carbon values, thermal properties, and market-verified costs as of early 2026. Materials are filtered by ECBC climate zone, budget tier, and local availability.

Materials by Category

Category	Count	Cost Range (Rs./m²)	Key Properties Tracked
Wall Materials	22	800 – 5,500	Embodied carbon (kg CO2/kg), thermal conductivity (k), density, life, availability zones, fire rating, water absorption, compressive strength
Roof Materials	15	250 – 3,750	Solar Reflectance Index (SRI), absorptivity, U-value, life, embodied carbon, wind resistance
Floor Materials	14	60 – 1,675	Thermal conductivity, density, recycled content %, life, maintenance interval
Window / Glazing	8	2,400 – 14,000	U-value (W/m²K), SHGC, visible light transmittance, frame U-value, life
Insulation	13	80 – 680	Thermal conductivity (k), density, renewable flag, fire rating, life

Carbon Ratings — A+ to E Scale

Rating	Embodied Carbon Range	Example Materials
A+ (Best)	< 0.05 kg CO2/kg	Adobe/Mud, Stone Rubble, Laterite, Thatch, CSEB, Shahabad Limestone
A	0.05 – 0.15 kg CO2/kg	Fly Ash Brick, Bamboo, Jute/Coir insulation, Kota Stone, Recycled brick
B	0.15 – 0.25 kg CO2/kg	Hollow Concrete Block, Rat-Trap Bond, Ferro-cement, Precast RCC
C	0.25 – 0.50 kg CO2/kg	Burnt Clay Brick, AAC Block, RCC Slab, Ceramic Tile, Vitrified Tile
D	0.50 – 2.00 kg CO2/kg	All glazing types (aluminium and uPVC), Rock Wool
E (Worst)	> 2.00 kg CO2/kg	GI Sheet, EPS Thermocol, XPS Polystyrene, Foil Bubble Wrap

Locally Sustainable Materials in GreenNirman

Material	Region Available	Why Sustainable	Carbon vs Brick
Laterite Stone Block	Kerala, Karnataka coastal, Goa	Quarried locally, hardens with age, no firing	-92% carbon
Bamboo (structural)	Northeast, Western Ghats	Grows back in 3-5 years, carbon-sequestering	Carbon negative
Fly Ash Brick	Near thermal power plants (all states)	Uses industrial waste, 60% less carbon	-67% carbon
CSEB / Compressed Earth	Dry regions, rural India	Local soil, minimal processing, zero transport	-88% carbon
Rice Husk Ash Board	Punjab, Haryana, UP, WB	Agricultural waste from 90M tonnes/yr production	-79% carbon
Jute Fibre Batts	West Bengal, Assam, Bihar	India produces 70% of world's jute — abundant	-50% carbon
Coconut Coir Board	Kerala, Tamil Nadu, Karnataka	Coconut husk waste — zero-waste processing	-58% carbon
Mangalore Clay Tile	South India, coastal	Traditional fired clay — natural ventilation	-44% carbon vs RCC
Jack Arch Brick Roof	North and Central India	30-40% less steel than RCC slab	-35% carbon vs RCC
Adobe / Mud Wall	Rajasthan, arid zones	Near-zero carbon — traditional earthen construction	-92% carbon

7. Calculation Methodology

GreenNirman uses a seven-parameter scoring system grounded in Indian building standards. All calculations run client-side in the browser using embedded data — no server, no internet required at assessment time.

Seven Sustainability Parameters

Parameter	Weight	What It Measures	Indian Standard
Embodied Carbon	30%	CO2 emitted in manufacturing, processing and transporting all building materials	ICE Database v3, Reddy & Jagadish 2003, BMTPC 2021
Thermal Performance	25%	U-values of wall, roof, window vs ECBC limits; OTTV and RTTV compliance	ECBC 2017 Section 4 & 5, IS 3792
Material Availability	15%	Local availability score of selected materials — reduces transport carbon	BMTPC availability zones, NBC 2016
Cost Efficiency	10%	User material cost vs conventional baseline — incentivises affordable green choices	CPWD DSR 2024 baseline costs
Material Durability	10%	Average service life of selected materials — longer life = lower lifecycle carbon	CPWD maintenance schedules, IS standards
Renewable & Recycled	5%	Fraction of renewable or recycled materials selected	GRIHA criterion 19-21, NBC 2016
Water Management	5%	Rainwater harvesting adoption, district water scarcity risk	IS 15797, CGWB norms, NBC 2016 Part 9

OTTV/RTTV Calculation — District-Specific

The Overall Thermal Transfer Value (OTTV) for walls and Roof Thermal Transfer Value (RTTV) are calculated using the actual district climate data from NASA POWER — not generalised zone averages. This means a building in Jaisalmer and a building in Jodhpur (both Zone 1) receive different thermal analyses based on their real measured solar radiation and temperature profiles.

- OTTV formula: Uw × (1-WWR) × TDeq_wall + Uw_n × WWR × dT_win + SC × WWR × SF × orientation_factor
- Direction-specific WWR applied: North, South, East, West individually
- RTTV: Ur × TDeq_roof + Ir × absorptivity × SF_roof
- Solar Factors (SF) computed from district monthly irradiance × 30 days × 1000

- COP 3.5 assumed for cooling (BEE 5-star split AC, ISEER >= 5.0)
- Grid emission factor: 0.727 kg CO2/kWh (CEA version 20, FY2023-24)

GRIHA Star Rating Scale

GRIHA Points (out of 24)	Stars	Performance Level	Policy Significance
< 8 points	1 Star	Below ECBC baseline — non-compliant	Does not qualify for any green incentive
8 – 11 points	2 Stars	Meets minimum ECBC 2017 requirements	Minimum compliance for large buildings >500m²
12 – 15 points	3 Stars	Good — significantly better than baseline	Eligible for PMAY green housing subsidies
16 – 19 points	4 Stars	Very Good — strong performance	Eligible for BEE star label, state incentives
20 – 24 points	5 Stars	Excellent — net-zero direction	Eligible for GRIHA certification pathway

ECBC 2017 Compliance Limits — All 5 Zones

Compliance Check	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
Max Wall U-value (W/m²K)	0.44	0.44	0.44	0.44	0.30
Max Roof U-value (W/m²K)	0.33	0.33	0.33	0.33	0.20
Max Window U-value (W/m²K)	3.30	3.30	3.30	3.30	2.00
Max OTTV — Wall (W/m²)	45	50	45	40	N/A
Max RTTV — Roof (W/m²)	25	25	25	25	N/A
Eco Niwas RETV (W/m²)	15	15	15	15	N/A
Min Window-Floor Ratio	10%	10%	10%	10%	5%

8. Platform Features

Input Parameters — 22 Variables

Input	Range	Connected Output
District / Location	758 Indian districts with typeahead search	Zone, climate data, insights, smart recommendations
Built-up Area	10 – 200,000 m²	All energy, carbon, water, and cost calculations
Number of Floors	1 – 40 (G+39)	Per-floor area validation, structural score
Number of Occupants	1 – 5,000	Water demand, domestic hot water energy, IAQ score
Building Type	Residential / Commercial / Industrial / Institutional	Baseline comparisons, ECBC applicability thresholds
Budget Category	Affordable / Standard / Premium / Luxury	Filters material list to show only cost-appropriate options
Wall Material	22 options	Embodied carbon, U-value, OTTV, carbon rating
Roof Material	15 options	RTTV, SRI, solar reflectance, embodied carbon
Floor Material	14 options	Embodied carbon, thermal mass contribution
Window / Glazing	8 options	U-value, SHGC, daylight, solar gain
Insulation	13 options	Effective U-value improvement, thermal resistance
WWR per face N/S/E/W	5% – 60% each direction	Directional OTTV, solar gain, daylight adequacy
Building Orientation	N/S/E/W/Composite	Solar gain per facade, OTTV direction weighting
HVAC Type	Split AC / Central / Natural Vent / Hybrid	COP value, energy benchmark, passive design credit
Rainwater Harvesting	Yes / No	Water score, RWH sizing recommendation, bylaw compliance
Solar PV	Yes / No	Net energy balance, operational carbon offset
Night Cooling	Yes / No	Passive design score, cooling load reduction
Low VOC Materials	Yes / No	Indoor air quality score, occupant health metric
Skylights	Yes / No	Daylight adequacy, lighting energy reduction
Permeable Surface %	0% – 80%	Biodiversity score, stormwater runoff management
Tree Count	0 – 50	Biodiversity score, annual carbon sequestration

Input	Range	Connected Output
Green Roof %	0% – 100%	Biodiversity, stormwater retention, urban heat island

Output & Analysis

- Overall Sustainability Score (0-100) with GRIHA star rating (1-5 stars)
- Seven-parameter breakdown with visual scoring bars and ECBC pass/fail flags
- Annual energy savings in kWh and Rs./year vs ECBC baseline building
- 25-year lifecycle cost comparison — selected materials vs conventional baseline
- Embodied carbon (tCO2) and operational carbon (tCO2/year) quantified separately
- Equivalent trees planted metric for carbon communication
- Peak cooling load (kW) — useful for right-sizing HVAC systems
- Monthly energy breakdown — cooling vs heating vs lighting vs DHW
- Smart recommendations panel — district-specific, ranked by carbon/cost impact
- Alternative materials engine — shows greener options with score improvement delta
- Water demand analysis — daily litres + RWH tank sizing recommendation
- District insights panel — seismic, flood, soil, water risk, traditional architecture
- Compare Mode — side-by-side assessment of two different material combinations
- Professional PDF report — full multi-page assessment for submission to authorities

9. Budget-Aware Material Filtering

India's construction sector spans an enormous cost range — from EWS housing at Rs. 800/sqft to luxury projects at Rs. 10,000+/sqft. GreenNirman's budget filtering ensures that material recommendations are always economically relevant to the project at hand.

Budget Tier	Max Mat. Cost	Target Segment	GRIHA Target	Typical Materials
Affordable	< Rs. 2,500/m²	EWS/LIG housing, PMAY, rural self-help	2 stars	Fly Ash Brick, CSEB, IPS Floor, Jute insulation, Jack Arch roof
Standard	< Rs. 5,000/m²	MIG housing, typical urban residential	3 stars	AAC Block, Hollow Block, Vitrified Tile, Rock Wool insulation
Premium	< Rs. 9,000/m²	HIG residential, mid-grade commercial	4 stars	Cavity Brick, Cool Roof coating, Double Glazed uPVC, Bamboo Strip floor
Luxury	All materials	Premium residential, green-rated commercial	5 stars	Hempcrete, Triple Glazed uPVC, Marble, Hemp Wool insulation

This tiering ensures that a rural PMAY homeowner and a LEED-seeking developer both receive material recommendations appropriate to their economic context — without changing any other parameter.

10. Assessment Report — PDF Output

GreenNirman generates a professional multi-page PDF assessment report entirely in the browser — no server, no upload, no data sharing. The report can be submitted to urban local bodies, development authorities, banks, and government schemes.

Report Sections

Section	Contents
Cover Page	Project name, district, climate zone, built-up area, building type, assessment date
Executive Summary	Overall score, GRIHA stars, ECBC compliance status, 3 key metrics at a glance
Score Breakdown	All 7 parameters with individual scores, radar chart, and explanations
ECBC Compliance	Wall/Roof/Window U-values vs zone limits, OTTV/RTTV/RETV with pass/fail
Energy Analysis	Annual kWh savings, Rs./year savings, 25-year lifecycle cost, monthly profile
Carbon Footprint	Embodied CO2 (tCO2), operational CO2 (tCO2/yr), saved vs baseline, tree equivalent
Material Summary	All 5 selected materials with full properties — cost, carbon, U-value, life, rating
District Insights	Seismic zone, flood risk, soil type, water scarcity, construction cost context
Smart Recommendations	Top prioritised design actions with expected impact quantified
Methodology & Sources	Standards used, data sources, calculation basis, known limitations, version

Standards Referenced in Reports

- ECBC 2017 (Energy Conservation Building Code) — BEE, MoP, Government of India
- Eco Niwas Samhita 2021 (residential energy code) — BEE
- NBC 2016 Part 8 (Building Services) — BIS
- IS 1893:2016 Part 1 (Seismic design criteria) — BIS
- GRIHA Rating System v.2019 — TERI / MoHUA
- CPWD DSR 2023-24 (construction cost index) — Ministry of Housing & Urban Affairs
- NASA POWER climatological dataset (2000-2020) — NASA Langley Research Center
- CEA Grid Emission Factor v20 FY2023-24 — Central Electricity Authority

11. Contribution to India's Sustainability Goals

Alignment with National Targets

National Goal	GreenNirman's Contribution
Net Zero by 2070 (India NDC)	Quantifies operational carbon for every building — creates awareness at project level
45% emissions intensity reduction by 2030	Embodied + operational carbon scoring pushes designers toward low-carbon material choices
PMAY — Pradhan Mantri Awas Yojana	Affordable budget tier ensures EWS/LIG housing can access green building guidance
Smart Cities Mission	District-level data supports city-level green building planning and bylaw development
Jal Jeevan Mission / Water conservation	CGWB water scarcity data + RWH sizing promotes district-appropriate water management
National Action Plan on Climate Change	Building sector mission alignment — energy efficiency + carbon reduction in construction
BEE ECBC 2017 enforcement	Makes ECBC compliance checking accessible beyond large commercial projects

Why District-Level Precision Matters

India's 758 districts span 5 climate zones, 6 major soil types, 5 seismic risk levels, and groundwater conditions ranging from Very Low to Critical. Building the same house with the same materials in Leh and in Chennai would result in completely different energy performance, comfort, durability, and cost outcomes.

- A cool roof (SRI>100) saves 25-30°C of roof temperature in Jaisalmer — but is irrelevant in Shimla
- Laterite block construction is ideal in Mangaluru — but not available in Punjab
- RWH is critical in Chennai (groundwater Critical) — less urgent in Aizawl (water Low)
- Bamboo framing is seismically superior in Imphal (Zone V) — but not code-compliant in Zone II
- A building in Kutch (Seismic Zone V) needs fundamentally different structural design than the same building in Hyderabad (Zone II)

GreenNirman makes these district-specific decisions accessible to every architect, engineer, and homeowner — not just those who can afford specialist consultants.

Potential Scale of Impact

Scenario	Estimated Impact
If 1% of India's 11M annual new buildings used GreenNirman guidance	~110,000 buildings with optimised material choices annually
Average 20% embodied carbon reduction per building (100 tCO_2 saved)	9.2 million tCO_2 avoided per year at 1% adoption (2023 baseline)
Average 30% cooling energy reduction per building	~15 billion kWh saved annually at 1% adoption
Average Rs. 50,000 lifecycle cost saving per building	Rs. 5,500 crore in household savings at 1% adoption
RWH adoption in water-critical districts	Reduced municipal water demand in 180+ over-exploited districts

12. Technical Summary

Attribute	Detail
Platform	React 18 — single-file browser application, no backend server required
Version	March 2026
Districts covered	758 Indian districts across 36 states and UTs
Climate data	NASA POWER API — 21-year averages (2000-2020), 670 districts with real data
Materials in database	72 verified materials across 5 categories
Standards implemented	ECBC 2017, NBC 2016, IS 1893:2016, Eco Niwas Samhita 2021, GRIHA v2019
PDF generation	jsPDF — fully browser-based, no data upload required
Data sources	NASA POWER, BIS, NDMA, NBSS&LUP, CGWB 2022-23, CPWD DSR 2024, IndiaMART
Offline capability	Full functionality without internet — all data embedded
Input parameters	22 variables covering location, area, materials, passive design, and services
Scoring parameters	7 weighted parameters producing GRIHA-mapped sustainability score (0-100)
Energy calculation	Monthly OTTV/RTTV model + HDD/CDD with district-specific NASA climate data
Carbon calculation	Embodied (ICE v3/BMTPC data) + operational (CEA grid factor 0.727 kg CO2/kWh)

Known Limitations

- 88 districts use zone-average climate data — specific small/remote districts have less precise energy calculations
- Building bylaws (FAR, ground coverage, setbacks) not populated for most districts — requires local authority research
- Nearby material supplier database not included — generic zone-level availability data only
- Material costs are indicative ranges (CPWD DSR 2024 base) — actual market prices vary 5-15% by location and time
- MEP systems (plumbing carbon, electrical distribution) not included in carbon calculation
- No BIM integration — all inputs are manual entry
- Some embodied carbon values from Indian research papers (Reddy & Jagadish 2003) are dated — updated values used where available from BMTPC 2021

Data Sources: NASA POWER | BIS IS1893:2016 | NDMA | NBSS&LUP | CGWB 2022-23 | CPWD DSR 2024 | ECBC 2017 | NBC 2016 | CEA FY2023-24